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Bengali Multi-class Text Classification via Enhanced Contrastive Learning Techniques

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Abstract—Bengali, one of South Asia’s most frequently spoken languages, poses substantial difficulties in tasks such as sentiment analysis and other forms of text classification due to its intricate grammatical structure. This is not just vital for protecting mental health through precise sentiment analysis, but it also has broader ramifications in sectors where accurately discriminating between fine-grained meanings is critical. Improving classification methods to address these subtle distinctions is a timely necessity for advancing natural language processing in Bengali. Our study aims to advance the field of Bengali text classification by implementing Token-level Adversarial Contrastive Training (TACT) and Label-aware Contrastive (LCL) loss, leveraging contrastive learning methods. The two new losses distinguished fine-grained text better, compared to our previous findings on Contrastive Adversarial Training (CAT) and Supervised Contrastive Loss (SCL). For binary class classification, TACT reached an F1-score of 98% outperforming CAT and LCL and setting a new benchmark on the Rokomari Book Review (RBR) dataset. For multi-class classification, TACT achieved an F1-score of 91%, matching the current benchmark on the Bengali Hate Speech (BHS-M) dataset. Furthermore, our custom Bengali multi-class text classification dataset, Daraz Product Review (DPR) further contributes to the field.

Index Terms—contrastive learning, bert, bengali text classification, token-level adversarial contrastive training, label-aware contrastive loss

I. INTRODUCTION

Bengali is a language that has evolved over more than 1,400 years, making it a complex language due to its non-static nature. Understanding and comprehending Bengali is difficult but necessary since it is the world’s seventh most widely spoken language in terms of total speakers. As it is the medium of expressing feelings for a large number of people, content in Bengali is widely available online, making it an intriguing research topic. Broad research has previously been conducted on Bengali-language web content, although much remains unexplored. The previously completed work also offers room for improvement.

In our efforts to go deeper into comprehending sentiment in Bengali text, we seek to address the limitations of existing methods. In Bengali, the phrase “এই গানটি মনে রাখতে ভুলবো না” (This song will not be forgotten easily) expresses a positive sentiment regarding a memorable encounter. However, the inclusion of “না” (not) in the context of “remembering” adds

complexity to the sentiment, introducing a twist to the text and potentially leaving a lasting favorable impression.

We aim to improve our comprehension of Bengali text classification by accounting for its complexities and capturing the various subtleties that contribute to the overall sentiment expressed. Through this technique, we hope to provide a more thorough and nuanced interpretation of sentiment, allowing for a better comprehension of the complexity inherent in Bengali language expression. This study follows up on our previous work [1] on Bengali social analysis.

Our approach combines the “push and pull” strategy, known as contrastive learning, with a pre-trained Bengali BERT model. This approach is one of the largely unexplored approaches with significant potential. The goal is to accurately capture the nuanced semantic choices embedded within sentiment-laden expressions with the help of Contrastive Learning.

The main contributions of this paper can be summarized as follows:

- We created a Bengali multi-class classification dataset for categorizing public text reviews on products.
- We have experimented with Token-level Adversarial Contrastive Training and Label-aware Contrastive Loss on Bengali text classification.
- Our TACT implementation either suppressed or matched the current benchmark performance for Bengali binary and multi-class text classification tasks.

II. RELATED WORKS

A. Bengali Text Classification

[2] introduces the Bengali Hate Speech Dataset (BHSD), a text-based dataset that identifies hate speech, collected from YouTube and Facebook comment sections. Their SVM model performed the highest at 87.5, while Random Forest and Naive Bayes showed accuracy of 70.1 and 52.20, respectively. [3] found that the transformer-based models, especially XLM-R, significantly outperformed the other models with an F1-score of 88.95%, which was 4.78% better than LSTM on their custom Bangla Book Review sentiment classification dataset. [4] presents a new dataset for Bengali text classification with 38

class labels. On the Bengali news classification dataset, their XLM-RoBERTa-Large model with Softmax classifier achieved an F1-score of 0.95. On their proprietary dataset, FastText with SVC received an F1-score of 0.92. [5] improves BERT-based model fine-tuning through semi-supervised learning on unlabeled data. The GAN-Bangla-BERT model obtained 85.4% accuracy on the Bengali Hate Speech dataset with 64 labeled samples, surpassing Bangla-BERT-Base by 29% and Bangla-Electra by 45.5%. They achieved 68.2% accuracy on the BanFakeNews dataset. [6] extended the Bengali Hate Speech Dataset and proposed an explainable approach for hate speech detection in Bengali. Their ensemble of different transformer-based architectures (Bangla BERT-base, mBERT-uncased, XLM-RoBERTa) achieved an F1-score of 88%. Our previous work [1], explores the use of SCL and CAT methods for Bengali language tasks, comparing them to traditional Cross-Entropy (CE) loss. Our BanglaBERT with SCL and CAT, achieving 93% accuracy for the BHS-B dataset. For the RBR dataset, BanglaBERT with CAT obtained 97%, outperforming the previous state-of-the-art.

B. Contrastive Learning Based Text Classification

[7] proposes distantly supervised contrastive loss (LDCL) and corpus-aware contrastive loss (CCL) for learning socio-pragmatic meaning in social media that does not require task-specific labels. It uses corpus-level knowledge to indirectly encode inter-class relationships. [8] leverages both text and audio modalities to capture offensive information, in addition to the mechanism of multi-task contrastive learning. [9] proposes contrastive adversarial training method to regularize the fine-tuning of Transformer-based encoders for text classification. During fine-tuning adversarial examples are generated which along with the clean texts are used to teach the model to work well on unseen data with the help of contrastive learning. [10] shows that contrastive adversarial training as a regularizer improves the performance of the model, which is analyzed using explainable AI. [11] uses CL to detect implicit hate speech using two positive sampling strategies. They proposed combined contrastive loss with cross-entropy loss to train generalizable implicit hate speech detectors.

III. DATASET

In our research, we used datasets of various sizes to evaluate how CL approaches perform across a range of data volumes, as well as their scalability and robustness. These datasets are as follows: Bengali Hate Speech - Binary (BHS-B) [2], Rokomari Book Review (RBR) [3], Bengali Hate Speech - Multiclass (BHS-M) [12], and Daraz Product Review (DPR). For DPR, 4004 samples have been used for experimental purposes. The RBR dataset consists of 1,445 instances among which 972 belong to the positive label class and 473 belong to belong to the negative class. Among the 30k posts for the BHS-B dataset, 10k are hateful and 20k are normal. Class wise data distribution for BHS-M¹ dataset along with examples can be found in Table I.

¹github.com/rezacsedu/Bengali-Hate-Speech-Dataset

TABLE I
CLASS-WISE DISTRIBUTION WITH EXAMPLES FOR BHS-M DATASET

Class	Values	Example
Political	814	ডঃ কামাল হোসেন ভারতের দালাল হয়ে ব্রিএনপিকে ধ্বংস করতে এসেছে। <i>Dr. Kamal Hussain has come to destroy BNP as an agent of India.</i>
Personal	2189	ফাপর দিয়া নিজেকে ওয়াদ ভাবা লোক ইনি <i>He is a man who thinks himself a master</i>
Geopolitical	1738	তারপরও নরেন্দ্র মোদী মহাভারতে সিংহ পুরুষ। <i>Still, Narendra Modi is lion like man in greater india.</i>
Religious	957	প্রিয় সকল মুসলিম ভাইদের বলছি এই সমস্ত মালাউন শয়তান থেকে সাবধান <i>Please all Muslim brothers beware of these evil devils</i>
Total	5698	

Daraz Product Review (DPR): For our custom text classification dataset, we employed manual data collection methods as well as web scraping to gather our corpus of data. We removed all input sequences from the web scraping that weren't in the Bengali language. The reviews in our dataset are classified into three labels: Negative, Positive, and Neutral. With two annotators participating in the annotation process, Cohen's kappa statistic was used to assess the inter-annotator agreement. The outcome was a Cohen's kappa value of 0.65 and a percentage of agreement of 97.26%. The dataset can be found in Github².

TABLE II
CLASS-WISE DISTRIBUTION WITH EXAMPLES FOR DARAZ PRODUCT REVIEW DATASET

Class	Values	Example
Negative	1613	খুব বাজে একটা পন্য <i>A very bad product</i>
Positive	3052	আলহামদুলিল্লাহ ভালো প্রডাক্ট এবং ভালো ভাবে পেয়েছি। <i>Praise be to God good product and well received.</i>
Neutral	649	খারাপ না <i>Not bad</i>
Total	5314	

Table III illustrates how annotators assigned the positive, negative, and neutral classes.

We split each dataset into training, validation, and test sets using a 70:15:15 ratio as outlined in Table IV.

IV. METHODOLOGY

Contrastive Learning is a learning method that helps to extract a meaningful representation by minimizing the distance between samples from the same classes and maximizing the distance between samples from different classes. CL helps in text classification by pulling together the samples from the same class and pushing apart the samples belonging to different class.

The BanglaBERT tokenizer prepares input text by segmenting it into tokens and adding special tokens for model compatibility, padding, and truncating sequences to fit the model's length limits. The model then generates contextualized token embeddings, which are used for classification, with the final hidden state of the [CLS] token providing the input for a softmax classifier.

²github.com/saymon005/Daraz-Product-Review-Classification

TABLE III
CLASS-WISE ANNOTATION RATIONAL FOR DARAZ PRODUCT REVIEW DATASET

	Positive Example	Neutral Example	Negative Example
Text	প্যাকিজিং সুন্দর ছিলো। অফারে কম টাকায় দেওয়ার জন্য ধন্যবাদ ইনফিনিট।	ভালই চলছে। তবে বেশি ঘন কফি হলে লোড নিতে পারেনা।	ভাবছিলাম যে ভালো প্রডাক্ট পাব কিন্তু পেলাম একটা ছেরা প্রডাক্ট
Translation	Packaging was beautiful. Thanks to Infinix for offering at a lower price.	It's going well. However, it can not handle heavy coffee load.	I thought I would get a good product, but instead received a torn product.
Annotation Rationale	Presence of words conveying appreciation, satisfaction, or praise. Here, phrases like “সুন্দর ছিলো” (was beautiful) and “ধন্যবাদ” (thanks) indicate a positive sentiment towards the packaging and the provided offer.	Lacking clear expression of strong positive or negative sentiment, containing emotionless and factual statements. Here, Phrases like “চলছে” (going well) and “বেশি ঘন কফি হলে লোড নিতে পারেনা” (can not handle heavy coffee load.) convey a factual description without strong emotional leaning.	Presence of words conveying disappointment, dissatisfaction, or criticism. Here, the phrase “ছেরা প্রডাক্ট” (torn product) indicates a negative sentiment towards the received product, suggesting dissatisfaction with its quality.

TABLE IV
DATASET SPLIT

Dataset	Total	Train(70%)	Test(15%)	Eval.(15%)
BHS-B [2]	30,000	21,000	4,500	4,500
RBR [3]	1,445	1,011	217	217
BHS-M [12]	5698	3988	855	855
DPR (ours)	4004	2802	601	601

The weights obtained from weighting are incorporated with NTXent loss to train the model.

$$\mathcal{L}_i = \sum_{j \in P_i} \log \left(\frac{w_{i,y_i} \cdot e^{\text{sim}(z_i, z_j)/\tau}}{\sum_{k=1}^{2N} l_{i \neq k} w_{i,y_k} \cdot e^{\text{sim}(z_i, z_k)/\tau}} \right) \quad (2)$$

$$\mathcal{L}_f = \sum_{i=1}^{2N} \frac{-L_i}{P_i} \quad (3)$$

A. Label-Aware Contrastive Loss (LCL):

In [13], a contrast loss known as LCL is introduced and proven effective in Arabic natural language processing (NLP) tasks [14]. This loss function considers the link between labels. For applications requiring fine-grained classification, this loss is helpful.

Fine-grained classification is the classification task of distinguishing closely related classes. The differences between these categories are very small. For example, there might be a sentiment analysis task in NLP that involves distinguishing between closely related categories, namely “positive” and “very positive.” Both classes have examples with almost the same meaning. In this case, the task of fine-grained classification is to classify highly mixed emotion pairs compared to classifying “positive” and “negative” class samples.

According to [13], when tackling fine-grained classification issues, employ a dual-model method. This technique adds a weighted network in addition to the primary encoder. Discovering the connections between labels is the job of weighted networks. A weighted network is responsible for boosting the confusable label X ’s weight in relation to other X labels. We use three losses to optimize the encoder and weighted network. Cross-entropy loss L_w is used to optimize the weighted network, and a linear combination of L_f and the cross-entropy loss L_e is used to optimize the encoder network’s output.

The prediction probabilities obtained from the softmax layer of the weighting network help to compute the probability of a sample belonging to a class c ,

$$w_{i,c} = \frac{e^{z_{i,c}}}{\sum_{k=1}^C e^{z_{i,k}}} \quad (1)$$

Here, C is the number of classes. $w_{i,c}$ denotes the probability of sample x_i belonging to class c .

Here, w_{i,y_k} denotes the relationship between a label y_k and an input sample x_i .

In our work, we use dropout-based data augmentation. The dropout-based data augmentation is used to produce an equivalent representation z_j , for the representation z_i . Both the representations z_i and z_j belong to the same class. For a confusable sample, the weighting network will assign higher weights to classes that are closely related to the sample. This high weighting score is used with the NTXent loss, which helps the model to learn better patterns to differentiate between confusable samples. The final loss is,

$$\mathcal{L}_{LCL} = (1 - \lambda)(\mathcal{L}_E + \mathcal{L}_w) + \lambda \mathcal{L}_f \quad (4)$$

B. Token-level Adversarial Contrastive Training (TACT):

Token-Level Adversarial Contrastive Training is a novel technique published in [15], which aims to improve the resilience and discriminative capacities of neural network models in NLP applications. The method was further experimented in [14]. Tokenization, the process of breaking up input text data into discrete units like words or subwords, is the first step in the TACT process. In the neural network architecture, each token is first represented as a vector. Traditionally, NLP approaches incorporate perturbations in the model’s input by perturbing complete word embedding matrices. Token representations are immediately adjusted by TACT, which adopts an alternative strategy. This makes it possible to precisely alter the representation of every token, which makes it easier to gather complex contextual data.

TACT minimizes a composite loss function in order to optimize the model parameters during training. This loss function combines contrastive features, like InfoNCE loss, to promote the learning of unique token representations, with

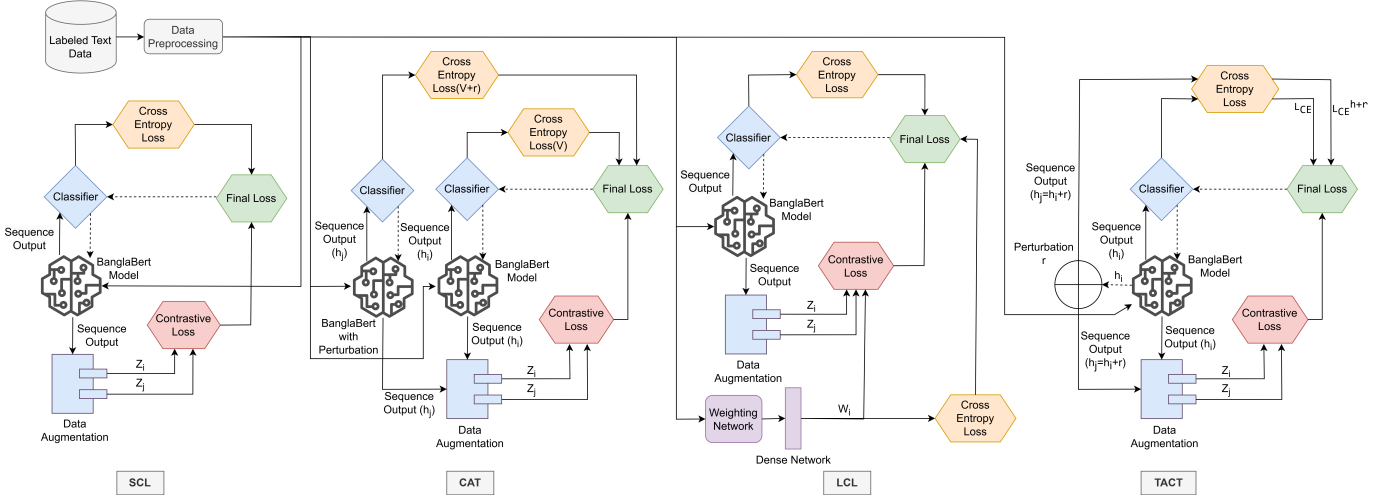


Fig. 1. Architecture for SCL, CAT, LCL and TACT.

supervised components, like cross-entropy loss for precise predictions. TACT's primary novelty is the way it perturbs token representations directly. To do this, gradients of the loss function with respect to each token representation must be computed, and the representations must then be modified correspondingly. Mathematically, the perturbation r for a token representation h_i is calculated as:

$$r = -\epsilon \frac{\nabla_{h_i} L(f(x_i, y_i))}{\|\nabla_{h_i} L(f(x_i, y_i))\|_2} \quad (5)$$

Where ϵ is a small perturbation factor, $\nabla_{h_i} L(f(x_i, y_i))$ represents the gradient of the loss function with respect to h_i . The perturbed token representation h_j is obtained by adding this perturbation to the original representation:

$$h_j = h_i + r \quad (6)$$

Similar to traditional methods, both the original token representation h_i and the perturbed representation h_j are passed through a non-linear projection layer. This layer helps the model capture complex relationships between tokens. The obtained representations are then used to train the model to minimize the InfoNCE (Info Normalized Cross-Entropy) loss, denoted as $\mathcal{L}_{InfoNCE}$.

The final loss function for this variant of CAT denoted as $\mathcal{L}_{TACT}$, combines the traditional cross-entropy loss $\mathcal{L}_{CE}$ and the $\mathcal{L}_{InfoNCE}$ loss, with a balancing parameter λ :

$$\mathcal{L}_{TACT} = \frac{1-\lambda}{2} \cdot (\mathcal{L}_{CE} + \mathcal{L}_{CE}^{h+r}) + \lambda \cdot \mathcal{L}_{InfoNCE} \quad (7)$$

This approach allows the model to learn discriminative token representations directly, offering more precise adjustments that capture contextual nuances effectively. By integrating adversarial training principles at the token representation level, TACT presents a promising avenue for enhancing the robustness and generalization capabilities of NLP models.

V. EXPERIMENTAL SETUP

The experiments were performed with a maximum sequence length set to 128 for both the BanglaBERT (ELECTRA)

and BanglaBERT-Base (BERT) models. We employed AdamW optimizer with weight decay of 0.01 and a linear learning rate scheduler with 0.05 warmup proportion. The regularization parameters λ and τ were given values of 0.5 and 0.3, respectively. Batch sizes of 16 and 32 were used, with varied learning rates of 0.0005, 0.00005, and 0.00001, as well as epochs of 10, 20, and 30. Performance was evaluated using F1-score, as we were working with imbalanced datasets. Early stopping was enabled, with patience set to 5 epochs.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

A. Binary Classification

As evident in Table V, TACT perform better than LCL for both pretrained Bert and Electra models. Overall, TACT outperformed LCL by 1% in some cases as TACT is a variant of CAT that uses token-wise perturbation instead of perturbing the whole embedding matrix, so it preserves the context of each token, which may be the same words but used in different sentences with a different context. Here, we used micro F1-score as in classification tasks for which every test case is guaranteed to be assigned to exactly one class, micro F1-score is equivalent to accuracy.

TABLE V
F1-SCORES FOR TACT, LCL ON THE BHS-B, RBR BINARY CLASS DATASETS

Dataset	Batch Size	Learning Rate	Epoch	F1-score			
				BanglaBERT		BanglaBERT-Base	
				TACT	LCL	TACT	LCL
BHS-B	16	0.0005	10	0.47	0.40	0.44	0.40
		0.00005	10	0.92	0.92	0.93	0.93
		0.00001	30	0.93	0.91	0.93	0.88
	32	0.0005	30	0.50	0.48	0.50	0.50
		0.00005	10	0.93	0.93	0.92	0.90
		0.00001	20	0.93	0.92	0.92	0.90
RBR	16	0.0005	10	0.86	0.80	0.44	0.40
		0.00005	10	0.98	0.96	0.88	0.84
		0.00001	10	0.92	0.92	0.87	0.87
	32	0.0005	10	0.89	0.44	0.33	0.40
		0.00005	30	0.94	0.89	0.88	0.87
		0.00001	20	0.94	0.97	0.89	0.89

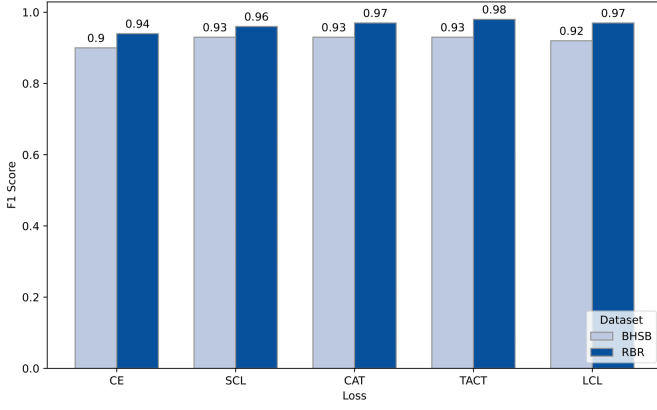


Fig. 2. F1-Scores for different losses on binary class datasets

Figure 2 compares TACT and LCL performance to losses from our previous study [1]. For the Binary class datasets, the RBR dataset got the better result for all the losses. For the BHS-B dataset, TACT beats all other methods. In summary, the performance of the CL methods beats CE in binary classification datasets.

B. Multi-class Classification

We have used two distinct Bengali text classification datasets for multi-class classification and implemented different methods involving different losses(CE, SCL, CAT, TACT, LCL). SCL, CAT, TACT, and LCL performed better than CE for both the BanglaBERT, BanglaBERT-Base for the two multi-class datasets as depicted in Table VI. These losses performed better as we used contrastive learning for our formulation.

For the BHS-M dataset, SCL and CAT outperformed CE (88%) by 2%, and similarly, TACT and LCL outperformed CE by 3% and 2%. The Figure 2 shows that the CL methods outperformed the CE method for the BHS-M dataset. Among the CL methods, TACT performed the best with an F1-score of 91%.

For the DPR dataset, our contrastive learning losses SCL, CAT, TACT, and LCL outperformed CE by 1%, 2%, 4%, and 1%, respectively. Among all the methods used, the highest F1-score was obtained for TACT. In Figure 3, the supremacy of the CL methods over the CE method still persists for the DPR dataset. The best-performing CL method is TACT with an F1-score of 77%. Between the two multi-class datasets we used, BHS-M gave us the best results for all the losses.

If we compare CAT with SCL, CAT outperforms SCL in some cases since CAT performs better than SCL in its adversarial training process. Adversarial training makes the model more robust for working with unseen data. In CAT, the adversarial examples are created for the purpose to make the model resistant when it sees unseen data in the real world. For example, for the DPR dataset, CAT beats SCL (74%) by 1% in F1-score, showing the advantage of adversarial training.

Compared to SCL, LCL has a slight edge in some cases. LCL uses a weighting network. The weighting network gives higher weights to classes that are closely related. Also, the

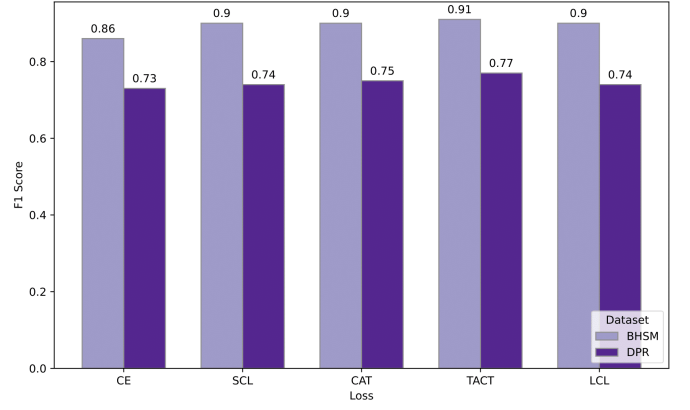


Fig. 3. F1-Scores for different losses on multi-class datasets

weighting network helps to reduce errors in classes of nominal categories by the help of weighting network.

TACT outperformed both CAT since TACT can be regarded as an improved version of CAT. The difference between TACT from CAT is that instead of perturbing the embedding matrix like CAT, TACT perturbs the tokens by keeping the context in consideration. The same word may have different meanings based on the use of the word in sentences and this context is preserved by TACT. As TACT preserves more contextual information than CAT, the performance of TACT is superior compared to the latter. For example, for the BHS-M dataset, the F1-score of TACT is 91% which is 1% higher than CAT. Similarly, for the DPR dataset, the margin of improvement was 2%.

For the DPR and BHS-M dataset, TACT outperformed all other losses. Though the performance of the CL methods is neck and neck, their outperforming the traditional CE method again proves the minimax nature of CL (minimizing the difference between the same classes and maximizing it between different classes), the fundamental and most important factor of our formulation is undoubtedly effective.

From the Figure 3, it can be seen that between the multi-class datasets, our methods performed better for the BHS-M dataset.

Throughout the study, we observed better performance for BanglaBERT model, compared to BanglaBERT Base, as BanglaBERT is an ELECTRA discriminator model and ELECTRA replaces the Masked Language Modeling (MLM) of BERT with Replaced Token Detection (RTD), which is proven to be more efficient and produces better results.

Regarding learning rate, 0.00005 and 0.00001 performs the best and while being indistinguishable from each other in terms of performance. In contrast, 0.0005 performs the worst, as a smaller learning rate has a smaller step size, so the minimum value of a parameter can be determined precisely and a larger learning rate, having larger step size does not perform better.

From Table VII, it is visible that CL-methods perform better than most of the existing models used for the dataset BHS-M. The statistical data in the Table VII indicates that the model “Bert+TACT” achieved the best performance overall on the

TABLE VI
F1-SCORES FOR CE, SCL, CAT, TACT, LCL ON THE BHS-M, DPR MULTI-CLASS DATASETS

Dataset	Batch Size	Learning Rate	Epoch	F1-score									
				BanglaBERT					BanglaBERT-Base				
				CE	SCL	CAT	TACT	LCL	CE	SCL	CAT	TACT	LCL
BHS-M	16	0.0005	10	0.39	0.44	0.40	0.40	0.40	0.39	0.40	0.40	0.39	0.40
		0.00005	30	0.84	0.90	0.90	0.91	0.90	0.80	0.83	0.82	0.87	0.82
		0.00001	10	0.84	0.86	0.86	0.85	0.85	0.80	0.87	0.85	0.85	0.84
	32	0.0005	10	0.40	0.43	0.40	0.44	0.40	0.40	0.42	0.43	0.32	0.40
		0.00005	10	0.88	0.89	0.88	0.89	0.87	0.86	0.87	0.86	0.87	0.86
		0.00001	20	0.87	0.86	0.87	0.88	0.87	0.85	0.86	0.85	0.86	0.85
DPR	16	0.0005	10	0.44	0.50	0.50	0.54	0.50	0.39	0.40	0.40	0.40	0.39
		0.00005	10	0.72	0.72	0.75	0.77	0.74	0.72	0.72	0.74	0.73	0.73
		0.00001	10	0.73	0.73	0.73	0.73	0.73	0.73	0.72	0.73	0.73	0.73
	32	0.0005	10	0.40	0.54	0.54	0.54	0.54	0.40	0.44	0.44	0.40	0.40
		0.00005	10	0.70	0.73	0.73	0.73	0.73	0.73	0.72	0.74	0.74	0.74
		0.00001	20	0.70	0.74	0.73	0.72	0.71	0.70	0.72	0.73	0.71	0.71

TABLE VII
PERFORMANCE COMPARISON FOR EXISTING MODELS ON BHS-B, RBR AND BHS-M DATASET

Dataset	Model	F1-score
BHS-B	BanglaBert+CAT [1]	0.93
	BanglaBert+TACT (ours)	0.93
RBR	BanglaBert+CAT [1]	0.97
	BanglaBert+TACT (ours)	0.98
BHS-M	Bangla BERT [6]	0.86
	XML-RoBERTa [6]	0.88
	GAN-BERT [5]	0.91
	BanglaBert+TACT (ours)	0.91

datasets, indicating the superiority of TACT over other CL-based methods.

VII. CONCLUSION

We evaluated the effectiveness of different contrastive learning methods, specifically Token-level Adversarial Contrastive Training and Label-aware Contrastive Loss, in Bengali text categorization. Our findings demonstrated that TACT beats top performers from our previous study, CAT and SCL, in binary and multi-class classification tasks, while LCL performed comparably to CAT. On the BHS-M dataset, our TACT implementation with the BanglaBERT model either matched or outperformed the current benchmark for multi-class classification. Our custom-made product review dataset will be beneficial in future studies focused on categorizing online consumer feedback. Future directions can be to generalize the model to work broadly for South Asian languages and introduce multi-modality, allowing us to identify emotions expressed through both text and image.

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